JITHENDRA PATHIRAJA

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linkedin.com/in/jithendra-pathiraja Portfolio

Moratuwa, Sri Lanka

Education

University of Moratuwa Moratuwa, Sri Lanka
B.Sc. Eng. (Hons.), Mechanical Engineering – First Class; 3.73/4.00 *2022 – 2026*
Dean's List: Semesters 2, 4 and 8

Maliyadeva College Kurunegala, Sri Lanka
GCE Advanced Level – Physical Science Stream; Z-score: 2.1052 *2012 – 2020*
GCE Ordinary Level (2017): 9A Passes

Wadugedara Primary College Kuliyapitiya, Sri Lanka
Primary Education *2007 – 2011*
Scholarship Examination: 184/200

Experience

Temporary Instructor Jul 2026 – Present
Department of Mechanical Engineering *University of Moratuwa*

Steering Systems Engineer May 2025 – Jun 2026
Team FalconE Racing *University of Moratuwa*

Team Member – Steering Aug 2024 – May 2025
Team FalconE Racing *University of Moratuwa*

Trainee Mechanical Engineer Dec 2024 – Jun 2025
Vega Innovations (Pvt) Ltd *Colombo, Sri Lanka*

Research Interests

Computational Fluid Dynamics (CFD), Computational Mechanics, Fluid Mechanics, Aerodynamics, Aerospace Structures, Aeroelasticity, Structural Design and Optimization, Finite Element Analysis (FEA), Numerical Simulation

Engineering Projects

Design and Development of a Hydrodynamic Tunnel for Experimental Fluid Dynamics

- Designed and developed a laboratory-scale hydrodynamic tunnel to generate controlled, low-turbulence flow for experimental fluid dynamics studies.
- Performed transient CFD simulations using ANSYS Fluent (SST k–ω model) to analyze flow behavior, pressure distribution, and optimize flow uniformity.
- Developed a comprehensive design framework for laboratory-scale hydrodynamic tunnels, including component design, engineering calculations, CAD modeling, and design optimization, followed by experimental validation.

Reusable Welding Jig for the Vega Street Bike Chassis

- Designed a reusable welding jig to ensure precise alignment and positioning of chassis members during fabrication, minimizing dimensional errors and improving repeatability.
- Manufactured using sheet metal and box section bars; components cut, drilled, and welded per SolidWorks manufacturing drawings.
- Aligned jig using a laser level across all axes prior to chassis assembly to ensure dimensional accuracy.

Rear Paddock Stand for the Vega Street Bike

- Designed and developed a rear paddock stand to elevate the motorcycle's rear wheel for maintenance operations, modeled as a full 3D CAD assembly in SolidWorks.

- Performed FEA using SolidWorks Simulation to verify structural integrity under applied hand force and motorcycle rear weight loading.
- Fabricated using mild steel hollow sections with MIG welding, positioned using a custom welding jig; finished with primer and paint for corrosion protection.

Front Suspension Dynamic Analysis – Vega Street Bike

- Developed a simplified dynamic model of the Vega Street Bike in MATLAB/Simulink to analyze front suspension response under bump excitation.
- Evaluated chassis displacement and studied the effect of suspension stiffness and damping parameters on ride behavior.
- Investigated critical damping characteristics to understand the influence of suspension tuning on vehicle stability and ride comfort.

Design of High-Speed Ambulance Craft

- Designed a high-speed ambulance vessel using regression-based analysis of existing craft data.
- Established relationships between principal dimensions to ensure stability and hydrodynamic performance.
- Selected optimal vessel dimensions based on statistical modeling and design constraints.

Steering System Optimization – Falcon E2 Formula Student Car

- Designed and developed an improved steering mount to enhance rigidity under high steering loads.
- Developed a custom jig for precise alignment of bearing housings during fabrication.
- Integrated and optimized steering system to improve performance and reliability.

Quick Jack System – Falcon E2 Formula Student Car

- Designed and fabricated a mechanical quick jack system for efficient lifting of the vehicle.
- Selected structural members to ensure strength, stability, and safe operation under load.
- Optimized design for ease of use and reliable load distribution.

Prosthetic Leg Fatigue Testing Mechanism

- Designed a mechanical system to simulate human gait for fatigue testing of prosthetic legs.
- Developed linkage mechanism and loading system to replicate realistic walking conditions.
- Modeled and fabricated prototype using SolidWorks and fabrication processes.

Numerical Analysis of NACA 4312 Airfoil

- Performed CFD analysis using ANSYS Fluent to evaluate lift, drag, and pressure distribution.
- Analyzed velocity contours to study aerodynamic performance under varying conditions.

Gearbox Design for Friction Washer

- Designed gearbox system and performed torque and power transmission calculations.
- Modeled assembly components using Solid Edge.

Reverse Engineering of Bicycle Air Pump

- Reverse engineered a bicycle air pump and developed CAD models of internal components.
- Conducted material identification using FTIR and microscopic analysis.

Smart Waste Management System

- Developed a CNN-based waste classification system using TensorFlow and OpenCV.
- Implemented automated sorting using Raspberry Pi and sensor integration.

Skills

Languages: English (professional proficiency), Sinhala (native proficiency)

Software:

- **3D Modeling:** SolidWorks, Solid Edge, AutoCAD, Rhinoceros 3D
- **Simulation:** SolidWorks Simulation, ANSYS, MATLAB (Simulink, Simscape Multibody), Orca3D, Maxsurf

Engineering: Mechanical Design, Design for Manufacturing (DFM), CAD Modeling, CFD Analysis, Structural Analysis, Mechanism Design, Steering System Design, Fabrication and Prototyping

Programming: Python, C++

Soft Skills: Communication, Leadership, Teamwork

Honors & Awards

IMechE SL Group Final Year Project Award Dec 2025
Institution of Mechanical Engineers (IMechE), Sri Lanka Group
– Awarded for outstanding performance, innovation, and teamwork in final year engineering project.
– Selected among top projects evaluated by IMechE Sri Lanka Group.

Professional Memberships

Institution of Mechanical Engineers (IMechE)
Affiliate Student Member | User Number: 80652142

Institution of Engineers Sri Lanka (IESL)
Student Member | Membership Number: S-30322

References

Dr. Indrajith D. Nissanka
Senior Lecturer, Department of Mechanical Engineering
Faculty of Engineering, University of Moratuwa
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Dr. Lihil Subasinghe
Senior Lecturer, Department of Mechanical Engineering
Faculty of Engineering, University of Moratuwa
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Mr. Chanaka Rathnayaka
CAD/CAM Lead, Vega Innovations (Pvt) Ltd
chanaka@vega.lk